

Theory of Two-level Tunneling Systems in Superconductors

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We develop a field theory formulation for the interaction of an ensemble of two-level tunneling systems (TLS) with the electronic states of a superconductor. Predictions for the impact of two-level tunneling systems on superconductivity are presented, including T_c and the spectrum of quasiparticle states for conventional BCS superconductors. We show that non-magnetic TLS impurities in conventional s-wave superconductors can act as pair-breaking or pair-enhancing defects depending on the level population of the distribution of TLS impurities. We present calculations of the enhancement of superconductivity, both T_c and the order parameter, for TLS defects in thermal equilibrium with the electrons and lattice. The scattering of quasiparticles by TLS impurities leads to sub-gap states below the bulk excitation gap, Δ , as well as resonances in the continuum above Δ . The energies and spectral weights of these states depend on the distribution of tunnel splittings, while the spectral weights are particularly sensitive to the level occupation of the TLS impurities. Under microwave excitation, or decoupling from the thermal bath, a nonequilibrium level population of the TLS distribution generates subgap quasiparticle states near the Fermi level which contribute to dissipation and thus degrade the performance of superconducting devices at low temperatures.

I. INTRODUCTION

One of the key sources of decoherence that limits the performance of superconducting devices for quantum computing and quantum sensing applications is two-level tunneling systems (TLS) in the oxides that coat the surfaces and interfaces of superconducting niobium and aluminum [1, 2]. Nb_2O_5 is known to limit the lifetime of microwave photons in superconducting RF cavities. Removal and prevention of re-growth of Nb_2O_5 leads to microwave photon lifetimes of $T_1 \simeq 2$ secs in Nb SRF cavity resonators [3]. Similarly, it was recently shown that removal of Nb_2O_5 from the Nb components of transmon qubits lead to significantly improved the qubit coherence times with $T_1 \gtrsim 500 \mu\text{sec}$ [4]. While removing surface oxide leads to improvement in microwave photon lifetimes in high-Q Nb SRF cavities and transmon coherence times, TLS disorder *embedded in* bulk Nb likely remains and can limit further improvement in the quality factor of Nb SRF cavities or coherence times of superconducting devices. Evidence of residual TLS disorder is inferred from the suppression of the quality factor at very low temperatures *after* the removal of Nb_2O_5 by heat treatment [5].

Known sources of TLS defects embedded in Nb are low concentrations of hydrogen (H), deuterium (D) and oxygen (O), that diffuse into Nb. Oxygen traps hydrogen and deuterium in the Nb unit cell. H and D tunnel between two symmetry related tetrahedral sites that are the local minima of a double-well potential energy profile for H/D trapped by O [5]. Evidence of atomic tunneling of H and D atoms in Nb is provided by heat capacity [6], ultra-sound [7], and inelastic neutron scattering [8, 9] measurements. Strain induced by the random spatial distribution of O-H and O-D impurities leads to a distribution of H/D tunnel splittings accessible by microwave photons [5, 6].

An additional source of decoherence for superconducting qubits is dissipation from quasiparticles under the nonequilibrium conditions of device operation. Losses from the *generation* of nonequilibrium quasiparticles are reported to be as important as other loss channels in transmon qubits when coherence times reach $200 \mu\text{sec}$ [10]. The effects of *static* disorder - magnetic and non-magnetic - on superconductivity have been studied extensively by many authors [11–14] since the development of the microscopic theory of superconductivity [15]. For recent developments related to microwave circuits and SRF cavities see Refs. [16–20]. There are a few studies of the effects of TLS impurities on superconductivity, including how the TLS impurities might mediate superconductivity [21], or enhance T_c in metallic glasses [22, 23], or act as Kondo-type defects [24]. In the opposite limit of isolated TLS impurities embedded in a superconductor, Yu and Granato considered the effect of pairing correlations on the magnitude of the tunnel splitting of an individual TLS, specifically a Hydrogen atom embedded in niobium [25]. They found that the tunnel splitting increases by $\approx 200 - 700 \text{ Hz}$ in the superconducting state compared to that in the normal state of Nb. This increase in tunnel splitting for a single H TLS impurity is very small compared to the width in the distribution of H tunnel splittings resulting from the strain induced by O impurities that trap H in Nb, even for the lowest OH concentrations ($c_O = 0.1\%$) measured [6].

In this report we develop the theory of embedded TLS impurities in conventional superconductors, such as

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Nb and Al, which in the clean limit are fully gapped BCS superconductors. We consider the regimes of weak to moderate disorder with $\hbar/2\pi\tau T_c \lesssim 1$ relevant to superconducting devices for qubits and sensors. Thus, we are neither in the regime of metallic glasses, nor the ultra-clean limit of isolated TLS impurities.

Our focus is on the effects of TLS disorder on the superconducting transition and the excitation spectrum. Unlike static nonmagnetic impurities, TLS disorder is a random distribution of dynamical impurities, which can be pair-breaking or pair-enhancing depending on the distribution of tunnel splittings and the level occupancies of the TLS impurities. One result of this study is that under equilibrium conditions we find a modest enhancement of T_c and the excitation gap, qualitatively consistent with earlier work on the effect of TLS in metallic glass superconductors [22, 23]. Another result of this study is that embedded TLS impurities generate sub-gap states below the clean limit excitation gap, Δ , as well as resonances in the continuum above Δ . The energies and spectral weights of these states depend on the distribution of tunnel splittings, with the spectral weights being particularly sensitive to the level occupations of the TLS impurities. Under the application of microwave power the level population of the TLS distribution can be driven out of equilibrium, in which case quasiparticle-TLS interactions are predominantly pair-breaking and generate a substantial density of subgap quasiparticle states. In particular, we show that *gapless superconductivity* can develop, and thus generate a substantial nonequilibrium population of quasiparticles which are a new channel for dissipation.

The rest of the paper is organized as follows: Sec. II describes the basic theoretical model defined by the Hamiltonian for a conventional BCS superconductor coupled to TLS impurities. We introduce Abrikosov's Fermion representation for the isospin operators describing each TLS impurity in Sec. III, then discuss Popov and Fedotov's method to eliminate unphysical states generated by the Fermion operators. The Fermion representation and Popov-Fedotov projection are central to the theoretical analysis that follows. In Sec. III we develop perturbation theory for the corrections to the Gorkov propagator generated by the quasiparticle-TLS interaction.

The distribution of TLS impurities is described by random variables for the position, orientation and tunnel splittings. Configuration averaging over the random distribution of TLS impurities is described in Sec. IIIB, and defines the TLS impurity self energies for quasiparticles and Cooper pairs discussed in Sec. IV. The functional form of these self energies are key results that allow us to calculate the effects of a random distribution of TLS impurities on the superconducting transition, order parameter and quasiparticle density of states (DOS). In Sec. V we report results for impact of TLS impurities on the equilibrium state of superconductivity, including the enhancement of T_c and the superconducting order parameter. We also report results for the density of states and spectra of sub-gap and above-gap resonance states resulting from multiple scattering by TLS impurities with equilibrium population levels. In Sec. VI we investigate the impact of TLS impurities with level occupations that are out-of-equilibrium with respect to the thermal bath of electron and phonon excitations. A key observation is that non-equilibrium TLS impurities are strong pair breakers generating a substantial density of low-energy sub-gap states that are likely sources of dissipation for disordered superconducting devices under microwave excitation.

II. THEORETICAL MODEL

The modern theory of superconductivity is based on a finite-temperature quantum field theoretical formulation for both thermodynamics and nonequilibrium transport and dynamics. The dynamics of TLS impurities is based on an isospin representation for the quantum state of the TLS impurities and their coupling to the electronic degrees of freedom. However, the isospin operators describing the dynamics of the TLS impurities do not obey the Wick theorem that is central to the traditional quantum field theory formalism, specifically the diagrammatic formalism used to define and classify the terms contributing to the electronic self energy resulting from the interaction with TLS impurities. We adopt the Fermion method introduced by Abrikosov [26] that allows us to represent the isospin operators in terms of local Fermion fields for the states of each TLS impurity, and the projection method of Popov and Fedotov [27] to eliminate unphysical states of the local Fermions. The representation of TLS impurities in terms of localized Fermions allows us to take advantage of standard quantum field theory methods to calculate the effects of TLS impurities on pairing correlations, T_c , and the quasiparticle spectrum of superconductors, as well as nonequilibrium properties such as the response to electromagnetic fields.

We introduce the Hamiltonian for a BCS superconductor coupled to a distribution of TLS impurity atoms whose local dynamics is defined by tunneling in a double-well potential. We start from the Hamiltonian

$$H = H_{\text{BCS}} + H_{\text{TLS}} + H_{\text{e-TLS}}, \quad \text{where} \quad (1)$$

$$H_{\text{BCS}} = \int d^3r \psi_{\alpha}^{\dagger}(\mathbf{r}) \left(-\frac{\hbar^2}{2m} \nabla^2 - \mu \right) \psi_{\alpha}(\mathbf{r}) - \int d^3r \left\{ \psi_{\alpha}^{\dagger}(\mathbf{r}) \Delta_{\alpha\beta}(\mathbf{r}) \psi_{\beta}^{\dagger}(\mathbf{r}) + \psi_{\alpha}(\mathbf{r}) \Delta_{\alpha\beta}^{\dagger}(\mathbf{r}) \psi_{\beta}(\mathbf{r}) \right\}, \quad (2)$$

is the mean-field BCS Hamiltonian defined in terms of the field operators, $\psi_{\alpha}(\mathbf{r})$, for the spin $s = \frac{1}{2}$ conduction electrons, and an attractive interaction for spin-singlet, s-wave pairing, $V_{\alpha\beta;\gamma\rho}(\mathbf{r}, \mathbf{r}') = -\frac{1}{4} g \delta(\mathbf{r} - \mathbf{r}')$

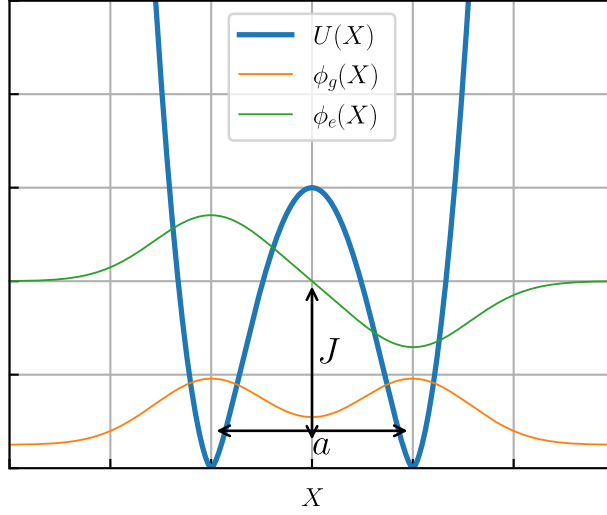


FIG. 1. Double well potential with the ground and excited state wave functions shown for the tunneling impurity. The tunnel splitting is J , and the distance between the two potential energy minima is a .

$\mathbf{r}') (is_y)_{\alpha\beta} (is_y)_{\gamma\rho}$, where s_y is the anti-symmetry Pauli spin matrix, and the $g > 0$ is the pairing interaction strength. The mean-field pairing energy is then $\Delta_{\alpha\beta}(\mathbf{r}) \equiv \Delta(\mathbf{r})(is_y)_{\alpha\beta}$, with $\Delta(\mathbf{r}) = \frac{1}{2} g (-is_y)_{\alpha\beta} \langle \psi_\alpha(\mathbf{r}) \psi_\beta(\mathbf{r}) \rangle$, where the condensate amplitude, $\langle \psi_\alpha(\mathbf{r}) \psi_\beta(\mathbf{r}) \rangle$, is the equal-time limit of Gorkov's anomalous propagator.

The second term in Eq. (1) is the Hamiltonian for the dynamics of N TLS impurities of mass M in a double-well potential, $U(\mathbf{X})$, such as that in Fig. 1,

$$H_{\text{TLS}} = \sum_{j=1}^N \left\{ \frac{|\mathbf{P}_j|^2}{2M} + U(\mathbf{X}_j) \right\}, \quad (3)$$

where $\mathbf{X}_j$ and $\mathbf{P}_j$ are canonically conjugate coordinates and momenta for the tunneling atom at the site $\mathbf{R}_j$. The dynamics of a single TLS impurity is governed by the Hamiltonian in Eq. (3) for $N = 1$, with the double-well potential, $U(\mathbf{X})$, of the form shown in Fig. 1. The two lowest energy states are the result of atomic tunneling through the barrier of the double-well potential between otherwise degenerate local minima separated by a distance $a \sim \text{\AA}$. These are the symmetric (ground), $|g\rangle$, and anti-symmetric (excited), $|e\rangle$ states with the wavefunctions, $\phi_{g,e}(X)$, as sketched in Fig. 1.

In what follows we retain only the two low-lying tunnel-split states of each impurity, i.e. we assume the high-energy states of the impurity Hamiltonian in Eq. (3), which are separated from the ground doublet by δE_{imp} , are inaccessible at low temperatures, $k_B T \ll \delta E_{\text{imp}}$, and low microwave photon energies, $\hbar\omega \ll \delta E_{\text{imp}}$. Thus, the Hamiltonian for the TLS impurities reduces to

$$H_{\text{TLS}} = \sum_j \frac{E_j}{2} \sigma_z(j), \quad (4)$$

where E_j is the energy difference between the excited and ground states of the j -th impurity, and $\sigma_z(j)$ is the diagonal Pauli matrix for the isospin representation of the j^{th} two-level tunneling atom. For identical, non-interacting TLS impurities in a symmetric double-well potential the energy splittings E_j are degenerate and equal to the tunnel splitting matrix element J . However, the random distribution of TLS impurities generates a strain field within the metal that generates asymmetry in the minimum energies, $\{\varepsilon_j | j = 1, \dots, N\}$, of the double-well potential for each impurity, as well as a random distribution of tunnel matrix elements, $\{J_j | j = 1, \dots, N\}$, leading to a distribution of energy level splittings,

$$E_j = \sqrt{J_j^2 + \varepsilon_j^2}. \quad (5)$$

The third term in Eq. (1) describes the interaction between the conduction electrons and the TLS impurities,

$$H_{\text{e-TLS}} = \sum_\alpha \int d^3r \psi_\alpha^\dagger(\mathbf{r}) \sum_j V(\mathbf{r} - \mathbf{X}_j) \psi_\alpha(\mathbf{r}), \quad (6)$$

where $V(\mathbf{r} - \mathbf{X}_j)$ is the interaction potential between an electron at $\mathbf{r}$ and the j^{th} TLS impurity with position operator $\mathbf{X}_j$. If the impurity positions are static, e.g. $M \rightarrow \infty$, then, $H_{\text{BCS}} + H_{\text{e-TLS}}$ defines the Hamiltonian